

Use of Floquet Buffer

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Part I

Sparsho: Use of Floquet Buffer (sublayer) in Thermodynamic neuron

1 Introduction

1.1 Current Problem in the Thermodynamic neuron architecture

In open-quantum systems we usually consider an effective picture to study a system as because the total entropy production rate of the system along with the bath is zero meaning:

$$\partial_t S_{vn}[\hat{\rho}_{tot}] = 0$$

so in order to define a non trivial entropy rate production which is done by defining an effective density matrix:

$$\hat{\rho}_{effective} = \hat{\rho}_s(t) \otimes_{\alpha} \hat{\tau}_{\beta_{\alpha}, \mu_{\alpha}}$$

where $\hat{\rho}_s(t)$ is essentially the pure states or the sum of the pure states of the total density matrix as $\hat{\rho}_{total}(t)$ meaning we can write this as :

$$\hat{\rho}_s(t) = -Tr\{\hat{\rho}_{tot}(t)\}$$

this is so-called the reduced density matrix

As we are considering a relative description the information lost is essentially given by then the quantum relative entropy is given by:

$$S[\hat{\rho}_{tot}][\hat{\rho}_{effective}] = S_{vn}[\hat{\rho}_s] + \sum_{\alpha} \beta_{\alpha} \langle \hat{H}_{\alpha} - \mu_{\alpha} \hat{N}_{\alpha} \rangle + \sum_{\alpha} \ln Z_{\alpha} - S_{vn}[\hat{\rho}_{tot}]$$

now if we differentiate both sides with time we essentially consider that the total entropy change of the system is zero and the partition function of the bath is also invariant in time so effectively we get that :

$$\Gamma = \partial_t S_{vn}[\hat{\rho}_s] + \frac{1}{k_{\beta}} \sum_{\alpha} \frac{J_{\alpha}}{T_{\alpha}}$$

this entropy is not ensured to be positive in case of non markovian systems as there might be a backflow of information from the bath. In the current architectures of thermodynamics based logic gates all of them work as autonomous machines meaning they donot need external clock pulses for logical output. So the issue here is that we cant control pretty much any of these

except for the intrinsic properties of the system. In modern CMOS based gates operate by maintaining high fidelity as because its terminals are isolated from each other intrinsically but that is not possible for qubit. Due to these disadvantages the thermodynamic not gate being analogous to perceptron is an interesting architecture but it here the question arises is there any way to develop robust thermodynamic gates without this information backflow or as the system exchanges information with the environment is there a way to feedback lost information for increasing accuracy?

1.2 Proposed Floquet Buffer architecture

In the original Markovian thermodynamic neuron, the computational subsystem is directly coupled to four thermal reservoirs that act as the input, output, reference, and modulator baths. Since the qubits are in continuous contact with these reservoirs, information transfer is not strictly unidirectional. Energy and information may flow from the baths to the qubits as well as from the qubits back into the baths. Consequently, part of the logical information encoded within the neuron can be lost to the environment before reaching the output channel. Furthermore, the logical state is inferred from observables associated with the output bath, making the readout process susceptible to thermal fluctuations, environmental noise, and measurement errors.

To overcome these limitations, we propose the insertion of periodically driven Floquet buffer layers between the neuron and selected reservoirs. Specifically, a Floquet buffer is introduced between the reference qubit and the reference bath, while another Floquet buffer is placed in the output branch between the collector subsystem and the output bath. The resulting architecture introduces an intermediate dynamical layer between the computational subsystem and its environment.

Rather than a direct coupling

$$\mathcal{C} \longleftrightarrow \mathcal{B},$$

where $\mathcal{C}$ denotes the computational subsystem and $\mathcal{B}$ denotes a thermal reservoir, the interaction is mediated through a periodically driven Floquet subsystem $\mathcal{F}$,

$$\mathcal{C} \longleftrightarrow \mathcal{F} \longleftrightarrow \mathcal{B}.$$

Here, $\mathcal{F}$ acts as an intermediate Floquet buffer whose Hamiltonian satisfies

$$H_F(t + T) = H_F(t),$$

with driving period $T = 2\pi/\Omega$. The periodic modulation generates a quasienergy structure that selectively controls the transfer of energy and information between the computational subsystem and the reservoir.

From a physical perspective, the Floquet buffer acts as a spectral and temporal filter. By engineering the driving frequency and modulation strength, unwanted transitions can be suppressed while desired energy-transfer channels are enhanced. As a result, the effective system-bath coupling becomes dynamically controllable. The Floquet layer therefore reduces information leakage into the reservoirs and simultaneously suppresses noise entering the computational subsystem from the environment.

In the output branch, the Floquet buffer can additionally implement a feedforward mechanism. Information extracted from the collector subsystem is first processed by the driven Floquet layer before reaching the output reservoir. The output signal therefore experiences an effective amplification and stabilization stage, improving the distinguishability of logical states and reducing readout errors.

Conceptually, this architecture is analogous to a CMOS voltage inverter. In a CMOS circuit, transistor networks provide isolation between the input and output while simultaneously enabling gain, noise suppression, and controlled signal propagation. Similarly, the Floquet buffers provide dynamical isolation between the thermodynamic neuron and its thermal environment while actively regulating energy transport through periodic driving. The periodically driven Floquet layers therefore function as quantum thermodynamic counterparts of the intermediate transistor stages found in conventional CMOS logic circuits. We have done a mathematical rigorous note in section ??.

2 What is a Floquet Subsystem?

Suppose there is Hamiltonian which satisfies this relation:

$$H(t + T) = H(t)$$

where T is the time period and is given by

$$T = \frac{2\pi}{\omega}$$

Such periodically driven systems are called Floquet Systems. Now the Floquet theory is the time analogous version of the Bloch theory which states that the solution of a Schrödinger equation given by:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

can be written as

$$|\psi_\alpha(t)\rangle = e^{-i\epsilon_\alpha t/\hbar} |\phi_\alpha(t)\rangle$$

where ϵ_α is the quasi energies.

2.1 Practical Examples of Floquet Engineering

A familiar example of Floquet engineering arises in frequency modulation (FM) communication systems. In FM transmission, the frequency of a high-frequency carrier signal is varied periodically according to an information-bearing signal. Mathematically, an FM signal can be expressed as

$$s(t) = A_c \cos [\omega_c t + \beta \sin(\omega_m t)]$$

where A_c is the carrier amplitude, ω_c is the carrier frequency, ω_m is the modulation frequency, and β is the modulation index. The sinusoidal term periodically modulates the phase and consequently the instantaneous frequency of the carrier signal.

For example, in FM radio broadcasting, an audio signal periodically modulates the carrier frequency, allowing information to be encoded and transmitted efficiently. This provides an intuitive analogy to Floquet engineering, where a physical system is subjected to a periodic drive. Just as periodic modulation in FM generates new spectral components around the carrier frequency, periodic driving in a Floquet system can generate new dynamical features and effective energy structures that are absent in the undriven system.

3 Floquet Buffer Extension to the Thermodynamic Neuron

Before creating any extension and complex architecture to the thermodynamic neuron we must first do a mathematical and numerical rigorous note on the original architecture and then create viable extensions from it.

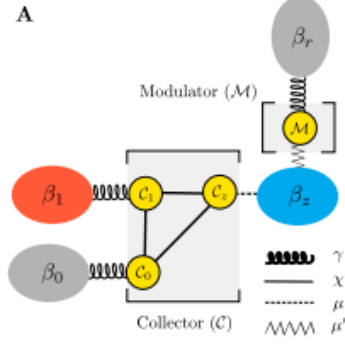


Figure 1: The Architecture of the thermodynamic Neuron
In the thermodynamic neuron

3.1 Original Thermodynamic Neuron architecture

In the thermodynamic neuron gate architecture there are 2 parts:

- **Collector:** This consists of 3 interacting qubits C_0, C_1, C_Z
- **Modulator:** This consists of a single Qubit C_M connecting the collector, through another bath B_r .

As shown in the Fig: 1 you can see that the collector module has 3 qubits amongst which 2 qubits C_0, C_1 are connected to baths B_0, B_1 amongst which the bath B_0 is the reference bath meaning having a constant temperature. The B_1 is the bath through which the information is encoded as input and passes through the collector. The 2 qubits are connected and directly interacting with the 3rd qubit C_z which is connected to a finite bath B_z which is a finite reservoir with heat capacity $\mathcal{C}$. We essentially take the logical output as a function of this bath in steady state. One might ask why is the modulator needed when the logical output is measured from B_z^∞ !!

Essentially without the modulator the thermodynamic not gate essentially acts as a linear amplifier not a robust logical gate. The ultimate goal of the collector is to create something called a virtual temperature which we will discuss later in the note and it tries to equilibrate the logical output to the virtual temperature making the input directly to the output i.e:

$$B_Z^\infty \approx B_v$$

with the modulator its job is to compete with the collector and constrain the possible values of B_Z^∞ acting as a filter to give the logical output.

3.1.1 The Collector architecture

Why need a collector? Consider a classical perceptron model given by :

$$y = \sum_i \omega_i x_i$$

before applying an activation function, the perceptron first computes the weighted sum of inputs. So the thermodynamic neuron is analogous to the perceptron model so instead of electrical voltage and the inputs are thermal reservoirs characterized by their inverse temperature.:

$$\beta_i = \frac{1}{K_B T_i}$$

Inside the collector contains 3 qubits C_0, C_1, C_z where C_0, C_1 are the input qubits with energy bandgap ϵ_1, ϵ_0 and the C_z is the output qubit with band gap ϵ_z .

Lets first do an numerical rigourous interpretation of the collector: So there are 2 input qubits C_1 and C_0 now thses qubits are connected 2 reservoirs B_0, B_1 amomgst whovht the bath connected to qubit 0 is a constant bath meaning the it acts a refrence , while thebath connected tot qubit 1 is through which tht e input is necoded into and sent to the output qubit. The thermal states of each baths are :

$$\tau_0 = \frac{e^{-\beta_0}}{Z_0}, \tau_1 = \frac{e^{-\beta_1}}{Z_1}$$

For a single qubit the excitation population number is given by :

$$\mathcal{H}_i = \epsilon_i |1\rangle \langle 1|$$

where the excited propbabailty is

$$\mathcal{P}_i(1) = \frac{e^{-\beta_i \epsilon_i}}{1 + \beta_i \epsilon_i}$$

and the ground state population is given by:

$$\mathcal{P}_i(0) = \frac{1}{1 + \beta_i \epsilon_i}$$

so the boltzman law states that:

$$\frac{\mathcal{P}(1)}{\mathcal{P}(0)} = e^{-\beta_i \epsilon_i}$$

Now for the combined system we do the same:

$$\tau_0 \otimes \tau_1$$

Now the basis states are

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

now the populations for each qubit is :

$$\mathcal{P}_{00} = p_0(0)p_1(0), \mathcal{P}_{01} = p_0(0)p_1(1), \mathcal{P}_{10} = p_0(1)p_1(0), \mathcal{P}_{11} = p_0(1)p_1(1),$$

In the above basis vector state of the system what it is done is that they have defined the states $|10\rangle, |01\rangle$ as the virtual qubits because unlike the other pure states they a form 2D subspace which can be seen in Fig ?? . So mathematically we define

$$|10\rangle = |1\rangle_v, |01\rangle = |0\rangle_v$$

Now calculating the virtual population of the virtual qubits

$$\mathcal{P}_v(1) = \mathcal{P}_{10}, \mathcal{P}_v(0) = \mathcal{P}_{01}$$

Now using the boltzmann ratio we get :

$$\frac{\mathcal{P}_v(0)}{\mathcal{P}_v(1)} = \frac{\mathcal{P}_0 \mathcal{P}_1}{\mathcal{P}_1 \mathcal{P}_0}$$

Using the Boltzmann relations

$$\frac{p_0(0)}{p_0(1)} = e^{-\beta_0 \epsilon_0},$$

and

$$\frac{p_1(0)}{p_1(1)} = e^{-\beta_1 \epsilon_1},$$

we obtain

$$\frac{p_1(1)}{p_1(0)} = e^{+\beta_1 \epsilon_1}.$$

Substituting into the population ratio yields

$$\frac{P_v(0)}{P_v(1)} = e^{-\beta_0 \epsilon_0} e^{+\beta_1 \epsilon_1}.$$

Hence,

$$\frac{P_v(0)}{P_v(1)} = e^{-(\beta_0 \epsilon_0 - \beta_1 \epsilon_1)}.$$

Now consider a genuine thermal qubit with energy gap E . Its thermal population ratio is

$$\frac{P(0)}{P(1)} = e^{-\beta E}.$$

The virtual subspace exhibits exactly the same functional form. We therefore define a virtual energy gap

$$E_v = \epsilon_0 - \epsilon_1,$$

and a virtual inverse temperature β_v through

$$\frac{P_v(0)}{P_v(1)} = e^{-\beta_v E_v}.$$

Comparing the exponents gives

$$\beta_v(\epsilon_0 - \epsilon_1) = \beta_0 \epsilon_0 - \beta_1 \epsilon_1.$$

Therefore,

$$\beta_v = \frac{\epsilon_0}{\epsilon_0 - \epsilon_1} \beta_0 - \frac{\epsilon_1}{\epsilon_0 - \epsilon_1} \beta_1$$

This quantity is called the virtual inverse temperature.

The terminology “virtual” is used because no physical thermal reservoir exists at temperature $T_v = 1/(k_B \beta_v)$. Rather, this effective temperature emerges from the population ratio

$$\frac{P_{01}}{P_{10}}.$$

The virtual qubit is therefore not a fundamental subsystem but an effective two-level system encoded within the composite Hilbert space of the two thermal qubits C_0 and C_1 .

To extract and utilize this virtual temperature, a third qubit C_z is introduced. Its energy gap is chosen to be resonant with the virtual energy gap:

$$\epsilon_z = \epsilon_0 - \epsilon_1 = E_v.$$

This C_z is connected to another bath B_z . So due to this creation of virtual temperature of the initial input qubits the output qubit temperature tries to equalize the virtual temperature or to be more precise the bath connected to the output qubit tries to equilibrate to this virtual temperature resulting in a linear response behaviour making it necessary for the presence of the Modulator.

3.1.2 Modulator

The Modulator is made of a single qubit $C_{\mathcal{M}}$ connected to the collector output bath β_Z and another reference bath β_r .

Part II

Abhishek: Periodically Driven Open-System Logic Gates

4 Introduction

4.1 What is thermodynamic computing?

Thermodynamic computing [1] proposes that the natural relaxation of a physical system coupled to heat baths can be harnessed to perform computation. In the quantum regime, autonomous thermal machines composed of a few interacting qubits can act as “thermodynamic neurons”: logical inputs are encoded in the temperatures of finite-size heat baths, and the logical output is read from the temperature of an auxiliary output bath once the system has settled into a nonequilibrium steady state (NESS) [2, 3]. The key appeal is that such devices operate without an external clock: computation is driven entirely by the flow of heat from hot to cold reservoirs.

4.2 Why add periodic driving?

All existing quantum thermodynamic neurons are *autonomous*, i.e., their Hamiltonians are time-independent. In this note we ask a natural question: can one improve the speed or accuracy of a thermodynamic neuron by adding a periodic driving field? This is a form of *Floquet engineering*: a periodically driven auxiliary system (the *Floquet buffer*) is inserted between the logical qubits and the physical baths. The buffer acts as a tunable effective environment whose spectral density can be shaped by the drive.

The price is twofold. First, the device is no longer autonomous; it requires a classical clock to maintain the drive and to read out the result stroboscopically. Second, an external work input is needed to power the drive. Nevertheless, the resulting architecture—a *periodically driven dissipative logic gate*—is an interesting candidate for low-energy computation.

4.3 Scope and structure of this note

This note is written as a self-contained pedagogical resource for a collaborative project involving a theoretical physicist, a mathematician working on tensor networks, and an MSc student with a background in engineering and coding. We assume familiarity with basic quantum mechanics (Hilbert spaces, operators, partial trace, von Neumann equation) and statistical mechanics (canonical ensemble, partition function). All more advanced concepts (GKSL equation, Floquet theory, chain mapping of baths, matrix-product states) are reviewed in the main text or in the appendices.

The note is organised as follows.

- Section 5 provides concise reviews of the GKSL master equation, Floquet theory, the chain representation of bosonic baths, matrix-product states (MPS), and the trace distance as a distinguishability measure.
- Section 6 describes the Floquet-buffer architecture in detail.

- Section 7 states and proves a theorem (Proposition 7.1) that in a certain weak-coupling limit the logical qubits obey a time-periodic GKSL equation.
- Section 8 explains how to go beyond weak coupling using chain mapping and MPS, and how to extract energy currents and work without exponentially large bond dimensions.
- Section 9 proposes a conjectured work–distinguishability tradeoff.
- Section 10 outlines a staged numerical programme.
- The appendices contain additional technical details.

5 Background material

Here we collect, in a deliberately compact form, the mathematical tools that are essential for the rest of the note. The reader may skip this section on first reading and return as needed.

5.1 GKSL (Lindblad) master equation

A Markovian open quantum system is described by a density matrix ρ evolving under the GKSL (Gorini-Kossakowski-Sudarshan-Lindblad) equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (1)$$

where H is the effective system Hamiltonian, L_k are jump operators, and $\gamma_k \geq 0$ are positive rates. The generator is trace-preserving and completely positive. If the rates satisfy local detailed balance with respect to a heat bath at inverse temperature β , the system relaxes to a Gibbs state $\rho_\beta \propto e^{-\beta H}$ (when there is only one bath) or to a NESS when several baths are present.

5.2 Floquet theory and Floquet-Lindblad equation

A periodically driven system has a Hamiltonian $H(t) = H(t + T)$ with $T = 2\pi/\Omega$. The time-evolution operator can be written in terms of Floquet modes $|\phi_n(t)\rangle$ and quasi-energies ϵ_n :

$$U(t) = \sum_n e^{-i\epsilon_n t} |\phi_n(t)\rangle \langle \phi_n(0)|, \quad (2)$$

where $|\phi_n(t + T)\rangle = |\phi_n(t)\rangle$ and $\epsilon_n \in [-\Omega/2, \Omega/2)$.

When the system is open and weakly coupled to one or more baths, a Floquet-Lindblad equation can be derived [4, 5]. The generator becomes time-periodic:

$$\dot{\rho}(t) = \mathcal{L}(t)\rho(t), \quad \mathcal{L}(t + T) = \mathcal{L}(t). \quad (3)$$

The jump operators are expressed in the Floquet basis, and the rates depend on the Fourier components of the bath correlation functions at frequencies shifted by multiples of Ω . A crucial feature is the appearance of sidebands: transitions can occur at energies $\epsilon_m - \epsilon_n + q\Omega$, $q \in \mathbb{Z}$, which allows the system to absorb or emit quanta of the drive while exchanging energy with the baths.

5.3 Chain mapping of bosonic baths

A heat bath is usually modelled as a collection of independent harmonic oscillators:

$$H_B = \int_0^\infty d\omega \omega a_\omega^\dagger a_\omega, \quad (4)$$

coupled to the system by $H_{SB} = A \otimes \int d\omega \sqrt{J(\omega)}(a_\omega + a_\omega^\dagger)$, where $J(\omega)$ is the spectral density. For numerical purposes it is convenient to map this continuum model to a semi-infinite one-dimensional chain with nearest-neighbour hopping [6, 7]. The transformation uses a sequence of orthogonal polynomials (e.g., Lanczos or Chebyshev) determined by $J(\omega)$. After the mapping, the Hamiltonian becomes

$$H_{\text{chain}} = \sum_{k=1}^\infty \omega_k a_k^\dagger a_k + \sum_{k=1}^\infty t_k (a_k^\dagger a_{k+1} + a_{k+1}^\dagger a_k), \quad (5)$$

where a_k ($k \geq 1$) are new bosonic operators, and the system couples only to the first site: $H_{SB} \propto A \otimes (a_1 + a_1^\dagger)$. The parameters ω_k and t_k are determined recursively from the spectral density. For an Ohmic bath $J(\omega) \propto \omega$ with a suitable cutoff, the coefficients t_k decay rapidly with k , so that a truncation to a finite chain of length N (typically $N \sim 100 - 200$) gives converged results for the reduced system dynamics.

5.4 Matrix-product states (MPS) and the TDVP

A one-dimensional quantum lattice system (such as the chain-mapped bath) can be efficiently represented by a matrix-product state:

$$|\psi\rangle = \sum_{\{s_i\}} C^{s_1} A^{s_2} \cdots A^{s_N} |s_1 s_2 \cdots s_N\rangle, \quad (6)$$

where C and A^{s_i} are matrices, and the bond dimension χ controls the amount of entanglement that can be captured. The time-dependent variational principle (TDVP) or Trotter-Suzuki decompositions allow one to evolve the MPS in real time under a Hamiltonian $H(t)$. The computational cost scales as $O(N\chi^3 d^2)$ per time step, where d is the local Hilbert space dimension.

For the chain-mapped bath at moderate temperature, the entanglement entropy between the system and the first few chain sites saturates at a finite value. Consequently, the required bond dimension χ does not grow with the chain length N ; values of $\chi \approx 30-50$ are typical for converged simulations.

5.5 Trace distance as a distinguishability measure

The trace distance between two density matrices ρ and σ is

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (7)$$

where $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$. It satisfies $0 \leq D(\rho, \sigma) \leq 1$ and is a metric on the space of states. Operationally, $D(\rho, \sigma)$ is equal to the maximum success probability of distinguishing ρ from σ with a single measurement. We will use it to quantify the logical distinguishability: for a binary logic gate, the two possible output states $\rho_{\text{out}}^{(0)}$ and $\rho_{\text{out}}^{(1)}$ are perfectly distinguishable when $D = 1$, and completely indistinguishable when $D = 0$.

6 Architecture of the Floquet-buffered logic gate

The physical device is sketched in Fig. 2. It consists of:

- **Logical qubits $\mathcal{S}$** , described by a time-independent Hamiltonian $H_{\mathcal{S}}$. For concreteness, in the numerical programme we will use three qubits forming a NOR gate as in [2].
- **Floquet buffer $\mathcal{B}_F$** , a finite quantum system with a time-periodic Hamiltonian $H_{\text{buf}}(t) = H_{\text{buf}}(t + T)$, $T = 2\pi/\Omega$. The buffer could be, for example, a harmonic oscillator with a modulated frequency or a driven two-level system.
- **System-buffer coupling $H_{S\text{buf}}$** , assumed time-independent and, for the weak-coupling theorem, weak.
- **Thermal baths**, each at a different temperature T_α , coupled only to the buffer via H_{buf,B_α} .

The logical input is encoded in the temperatures $\{T_\alpha\}$ of the baths, exactly as in the autonomous thermodynamic neuron. The output, however, is not read from an auxiliary bath; instead, we measure an observable of the output qubit stroboscopically at times $t_n = nT$, after a periodic steady state has been reached. This is a deliberate departure from strict autonomy: the device requires an external clock to synchronise the drive and the readout.

The reasons for this choice are twofold. First, in the autonomous model the output bath temperature depends on the qubit-bath coupling, which is absent in our architecture (the qubits couple only to the buffer). Second, stroboscopic readout is natural in Floquet systems and is used in many quantum computing platforms.

Because we accept a clock, the thermodynamics of the device is that of a periodically driven open system. The relevant energetic quantities are the cycle-averaged work $\overline{W}$ provided by the drive and the cycle-averaged heat currents $\overline{J}_\alpha$ into the baths.

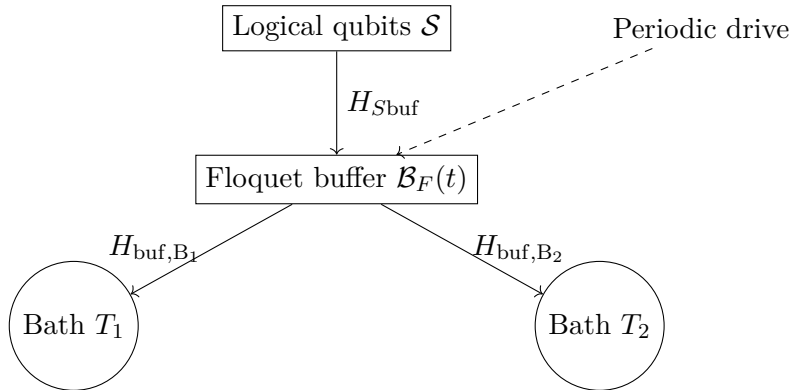


Figure 2: Periodically driven logic gate with Floquet buffer. The logical qubits interact only with the driven buffer; the buffer couples to thermal reservoirs. The device is not fully autonomous: a classical clock synchronises the drive and the stroboscopic readout.

7 Weak-coupling regime: periodic GKSL generator

Under certain idealised conditions, one can eliminate the buffer and the baths and obtain a closed description of the logical qubits alone.

7.1 Assumptions

We impose:

- (A1) **Weak buffer-bath coupling.** The interactions H_{buf,B_α} are sufficiently weak that the buffer, when decoupled from $\mathcal{S}$, relaxes to a unique periodic steady state $\rho_{\text{buf},ss}(t)$ under the Floquet-Lindblad equation. The generator of this buffer-only dynamics is $\mathcal{L}_{\text{buf}}(t)$.

- (A2) **Exponential decay of buffer correlations.** In that periodic steady state, the two-time correlation functions $\langle B_i(t)B_j(t-\tau) \rangle$ decay to zero on a characteristic time scale τ_c . More precisely, there exists τ_c such that $|C_{ij}(t, \tau)| < \varepsilon$ for $\tau > \tau_c$ for any chosen tolerance ε .
- (A3) **Weak system-buffer coupling and time-scale separation.** The coupling $H_{S\text{buf}}$ is much weaker than the buffer's internal relaxation scale, and the logical qubits evolve slowly compared to τ_c . This allows us to apply the Born–Markov approximation.
- (A4) **Secular approximation.** The Bohr frequencies (transition frequencies) of H_S are non-degenerate and well separated compared to the relaxation rates induced by the buffer.

7.2 Periodic GKSL generator

Proposition 7.1. *Under Assumptions (A1)–(A4), there exists a time-periodic generator $\mathcal{L}_{\text{eff}}(t)$ of the form*

$$\mathcal{L}_{\text{eff}}(t)\rho = -i[H_S + H_{LS}(t), \rho] + \sum_k \Gamma_k(t) \left(L_k(t)\rho L_k^\dagger(t) - \frac{1}{2}\{L_k^\dagger(t)L_k(t), \rho\} \right), \quad (8)$$

with $\mathcal{L}_{\text{eff}}(t+T) = \mathcal{L}_{\text{eff}}(t)$, such that the reduced state of the logical qubits obeys

$$\dot{\rho}_S(t) = \mathcal{L}_{\text{eff}}(t)\rho_S(t). \quad (9)$$

The jump operators $L_k(t)$ and rates $\Gamma_k(t)$ are obtained from the Fourier transforms of the buffer correlation functions evaluated at the Bohr frequencies of H_S , including sideband shifts by multiples of Ω .

Proof sketch. The derivation follows the standard Floquet–Born–Markov approach [4]. One starts from the Nakajima–Zwanzig equation with the buffer acting as a structured environment. The memory kernel is of second order in $H_{S\text{buf}}$. Because of (A2) the kernel decays on the time scale τ_c , and because of (A3) the system density matrix changes slowly on that scale; we can therefore make the Markov approximation and replace the memory integral by a time-local superoperator. The time dependence of this superoperator is inherited from the periodic buffer steady state. The secular approximation (A4) diagonalises the dynamics in the eigenbasis of H_S , yielding the GKSL form with time-periodic coefficients.

Practical limitations. The assumptions are strong. In particular, the weak-coupling condition (A3) may be hard to satisfy simultaneously with (A1) because weak buffer-bath coupling implies a long buffer correlation time, whereas a Markovian elimination requires the buffer correlations to decay fast *relative to the qubit dynamics*. Therefore Proposition 7.1 should be seen as identifying a parameter window in which a simple theoretical description exists, not as a statement about generic parameters.

A second issue is the dynamic Lamb shift. The Floquet buffer will induce time-dependent shifts $H_{LS}(t)$ in the qubit energy levels. The autonomous thermodynamic NOT gate relies on a precise resonance condition $\omega_W = \omega_X + \omega_Y$ among three qubit frequencies. The Floquet-induced Lamb shift will generically detune this resonance and destroy the logical operation. To mitigate this, we must actively adjust the bare qubit frequencies as functions of the drive amplitude, a procedure that is realistic in circuit-QED architectures but constitutes an additional control overhead.

For all these reasons, the main numerical programme of this project targets the regime *outside* the validity of Proposition 7.1, where no simple Lindbladian exists. Nevertheless, the weak-coupling analysis is a useful conceptual starting point.

8 Non-Markovian regime: chain mapping and MPS

To treat arbitrary coupling strengths and memory effects, we simulate the full system-buffer-baths compound as a closed many-body quantum system. We employ two established techniques: chain mapping of the baths (Sec. 5.3) and matrix-product state evolution (Sec. 5.4).

8.1 Chain-mapped Hamiltonian

Starting from the original bath model with a continuum of harmonic oscillators and a given spectral density $J(\omega)$, we apply a unitary transformation (Lanczos or orthogonal-polynomial mapping) that yields the chain Hamiltonian (5). The total Hamiltonian in the chain picture is

$$\begin{aligned} H_{\text{tot}}(t) = & H_S + H_{\text{buf}}(t) + H_{S\text{buf}} \\ & + \sum_{\alpha} \left[\sum_{k=1}^{N_{\alpha}} \omega_{\alpha k} a_{\alpha k}^{\dagger} a_{\alpha k} + \sum_{k=1}^{N_{\alpha}-1} t_{\alpha k} (a_{\alpha k}^{\dagger} a_{\alpha, k+1} + \text{h.c.}) \right] \\ & + \sum_{\alpha} V_{\text{buf}, \alpha} \otimes (a_{\alpha 1} + a_{\alpha 1}^{\dagger}). \end{aligned} \quad (10)$$

Here N_{α} is the chain length for bath α (chosen large enough to suppress finite-size effects), and $V_{\text{buf}, \alpha}$ is a buffer operator determined by the original buffer-bath coupling.

This Hamiltonian describes a one-dimensional lattice system. The logical qubits sit at the leftmost end, the buffer is next, and each bath forms a separate chain. The only interaction between qubits and baths is mediated by the buffer: the qubits couple to the buffer, and the buffer couples to the first site of each chain.

8.2 Energy current and work

Because the chain Hamiltonian is a closed system, thermodynamic observables can be defined exactly from the Heisenberg equations of motion.

The energy current from the logical qubits into the buffer is the time derivative of the qubit energy minus the contribution that would occur without the coupling:

$$J_{S \rightarrow \text{buf}}(t) = i \langle [H_S, H_{S\text{buf}}] \rangle. \quad (11)$$

Here $\langle \cdot \rangle = \text{Tr}(\rho_{\text{tot}}(t) \cdot)$, with ρ_{tot} the state of the full chain. Because H_S acts only on the qubits and $H_{S\text{buf}}$ acts on qubits and buffer, the commutator is a local operator with support on the qubits and the buffer alone. Its expectation value can therefore be computed directly from the MPS representation of the state without needing to reconstruct the full bath density matrix.

The power injected by the periodic drive is

$$P_{\text{drive}}(t) = \left\langle \frac{\partial H_{\text{buf}}(t)}{\partial t} \right\rangle. \quad (12)$$

This is also a local observable (it acts only on the buffer). The cycle-averaged work is

$$\overline{W} = \frac{1}{T} \int_0^T P_{\text{drive}}(t) dt. \quad (13)$$

The heat dumped into bath α can be obtained from the energy current flowing from the buffer into the first site of chain α , using the buffer-chain coupling. This current is again a local observable.

8.3 Bond-dimension control

A crucial practical question is whether the MPS simulation remains tractable. For a one-dimensional chain with nearest-neighbour couplings, the entanglement entropy of the state follows an area law: for a gapped system it saturates to a constant; for a critical (gapless) system it grows logarithmically with the chain length. The chain-mapped bath is gapless (the site frequencies ω_k extend to zero), but the coupling t_k decays rapidly with k , so that only the first few tens of sites are strongly entangled with the system.

In practice, for an Ohmic spectral density with a high-frequency cutoff and for temperatures $k_B T$ above the smallest energy scale of the chain, the bond dimension required for convergence is $\chi \approx 30\text{--}50$. This value does *not* increase when the chain length N is increased beyond the point where the first few sites have captured the relevant physics. The computational cost per time step then scales as $O(N\chi^3 d^2)$, where d is the local Hilbert space dimension. For a truncated Fock space with $d = 10\text{--}20$, this is manageable for the single-qubit buffer prototype (Phase 3 of the numerical programme) and, with tree-tensor-network (TTN) enhancements, for the three-qubit gate (Phase 4).

9 Conjectured work–distinguishability tradeoff

In the autonomous thermodynamic neuron, logical distinguishability is intrinsically linked to the temperature difference between the baths; increasing the distinguishability requires a larger temperature gradient, which implies larger entropy production. In our driven setting, we expect a similar tradeoff: improving the logical distinguishability beyond the undriven baseline should require a positive amount of work per cycle.

Formally, let

$$P_{\text{dist}} = \frac{1}{2} \left\| \rho_{\text{out}}^{(0)} - \rho_{\text{out}}^{(1)} \right\|_1 \quad (14)$$

be the trace-distance distinguishability of the output qubit states corresponding to the two logical inputs. Let $P_{\text{dist}}^{(0)}$ be this quantity when the buffer is undriven (i.e., the baseline autonomous gate, but with the same qubit readout), and let $\overline{W}$ be the cycle-averaged work when the buffer is driven.

Conjecture 9.1 (Work–distinguishability tradeoff). *There exists a monotonically increasing function f with $f(P_{\text{dist}}^{(0)}) = 0$ such that*

$$\overline{W} \geq f(P_{\text{dist}}). \quad (15)$$

In particular, any enhancement $P_{\text{dist}} > P_{\text{dist}}^{(0)}$ requires $\overline{W} > 0$.

This conjecture is motivated by known thermodynamic uncertainty relations (TURs) and speed-limit inequalities, which bound the precision of a process in terms of the entropy production or work. The precise functional form of f and its dependence on the bath temperatures are open questions that the numerical programme will explore.

10 Numerical programme

We propose a staged numerical investigation, designed to proceed from controlled to more challenging regimes.

Phase 1: Undriven baseline. Reproduce the autonomous NOT gate of [2] using QuTiP. However, instead of reading the output from a bath temperature, we read it from the reduced state of a designated output qubit. This gives us the undriven distinguishability $P_{\text{dist}}^{(0)}$ and the baseline steady-state heat currents.

Phase 2: Weak-coupling Floquet buffer (QuTiP). Implement a single qubit coupled to a driven harmonic oscillator (the buffer), which is weakly coupled to two Ohmic baths. Use QuTiP’s Floquet-Lindblad module to numerically verify Proposition 7.1. Explore how P_{dist} and $\bar{W}$ depend on drive amplitude and frequency. Check whether the conjecture holds in this regime.

Phase 3: MPS chain simulation (single qubit). Build the chain-mapped model of one qubit, one driven buffer, and two thermal chains using ITensor or MPSDynamics.jl. Initialise the chains in thermal states (via thermofield purification). Evolve in time until the periodic steady state is reached, then compute the energy current (11), the drive power (12), and the output qubit distinguishability. Investigate parameter regimes where the weak-coupling assumptions fail, and test the conjecture beyond weak coupling.

Phase 4: Three-qubit NOR gate. Extend the chain mapping to the full three-qubit NOR gate with a single Floquet buffer. Here the bond dimension may grow; if it becomes prohibitive, we will use tree tensor networks (TTNs) or systematic truncation of the least-coupled chain sites. The goal is to produce a phase diagram of P_{dist} as a function of drive amplitude and frequency, and to test the conjecture for a practical logic gate.

All code and data will be made publicly available to facilitate reproducibility.

11 Conclusion

We have presented a self-contained, pedagogical introduction to the Floquet-buffer architecture for periodically driven logic gates. The note reviews the necessary technical background, states a rigorous theorem in the weak-coupling limit, explains how to treat the non-Markovian regime with chain-mapped MPS, and formulates a clear conjecture to be tested numerically. The project is designed to utilise the complementary strengths of the collaboration: the physicist’s formal analysis, the mathematician’s tensor-network expertise, and the student’s numerical skills. This note is the definitive starting point for the work.

A Derivation of the GKSL equation from a microscopic model

For completeness we sketch the derivation of the GKSL equation (1) from a system-bath Hamiltonian. Consider a system S coupled to a bath B via $H_{SB} = \sum_i S_i \otimes B_i$. In the Born–Markov approximation, after tracing out the bath, one obtains the Redfield equation. The additional secular (rotating-wave) approximation diagonalises the dynamics in the eigenbasis of H_S and ensures complete positivity. The resulting jump operators are the eigenoperators $A(\omega)$ of the system Hamiltonian, with rates $\gamma(\omega) = \int e^{i\omega t} \langle B^\dagger(t) B(0) \rangle dt$. The detailed balance condition $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$ follows from the KMS condition of the bath. See [9] for a textbook treatment.

B Chain mapping: recurrence relations

Given a spectral density $J(\omega)$, the chain mapping is constructed by orthogonalising the polynomials in ω with respect to the measure $J(\omega)d\omega$. The recurrence coefficients ω_k and t_k are obtained from the continued fraction expansion of the bath correlation function. For the common case of an Ohmic bath with exponential cutoff, $J(\omega) = \eta\omega e^{-\omega/\omega_c}$, the coefficients can be computed analytically or numerically using the Lanczos algorithm [6].

C MPS representation of a thermal state

To initialise the chain in a thermal state at temperature T , one uses the thermofield (purification) approach. Each physical bosonic site is doubled into an ancilla site, and the initial state is the vacuum for a set of Bogoliubov-transformed modes. The resulting MPS bond dimension is directly related to the effective temperature; for moderate T , it remains small.

D Energy current operator: explicit form

Equation (11) follows from the Heisenberg equation for H_S :

$$\frac{dH_S}{dt} = i[H_{\text{tot}}, H_S] = i[H_S, H_{S\text{buf}}] + i[H_S, H_S] = i[H_S, H_{S\text{buf}}]. \quad (16)$$

The term $[H_S, H_S] = 0$, and H_S commutes with all bath and buffer-bath Hamiltonians because it acts on a different tensor factor. Hence the energy change of the qubits is solely due to the qubit-buffer interaction. Taking the expectation value gives the current.

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